



Lahore Garrison University

Department of Computer Sciences

Program/Semester: BSSE/6th

Course Code: SEE6603

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Course Title: Software Verification and Validation

Time Allowed: 60 minutes

Roll No: Fa-23/BSSE/148

Marks: 20

Validation Engineering and Compliance

Validation Checklist Engineering

Q1. Stakeholder Validation Checklist Design

1) Measurable Validation Criteria:

Stakeholder expectations & measurable validation rules:

Stakeholder Expectation	Measurable Validation Criteria
Secure Login	Login must use HTTPS + encrypted password hashing + 2FA support
Transaction Accuracy	Transaction success rate $\geq 99.9\%$ with correct balance updates
Session Timeout	User session expires after 5 minutes inactivity
Mobile Responsiveness	UI must adapt to screens $\geq 320\text{px}$ width
Audit Logging	All transactions must be logged with timestamp & user ID

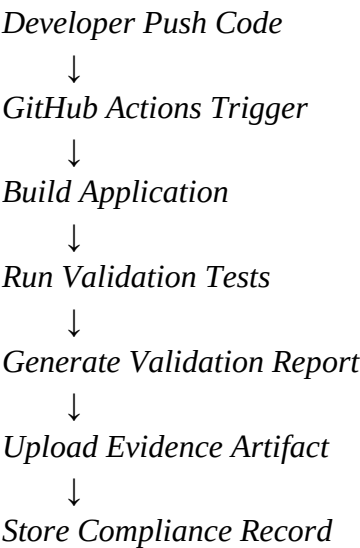
2) Structured Validation Checklist Table:

ID	Feature	Validation Item	Validation Method	Expected Result	Status	Evidence
VC-01	Login Security	HTTPS enabled	Browser inspection	Secure connection	Pending	Screenshot
VC-02	Login Security	Password encrypted	DB check	Hashed password	Pending	DB log
VC-03	Transactions	Accurate balance update	Functional Test	Correct balance	Pending	Test report
VC-04	Session Management	Auto logout	Idle test	Logout after 5 min	Pending	Video proof
VC-05	UI	Mobile responsive	Device test	Layout adjusts	Pending	Screenshot
VC-06	Audit Logging	Transaction logging	Log inspection	Entry recorded	Pending	Log file

Section B – Continuous Validation Evidence in CI Pipelines

Validation Checkpoints in CI Pipeline:

CI Pipeline me validation integrate karne ka flow:



Validation Checkpoints:

- *Code Build Validation*
- *Unit Tests Validation*
- *Security Validation*
- *Report Generation*
- *Evidence Archiving*

Section C – Security-Oriented Validation and Structural Vulnerability Examination**1. Role of Security Validation:**

Security validation ensures that software systems comply with security requirements by detecting vulnerabilities before deployment. It supports software verification and validation by confirming that applications are protected against unauthorized access, data breaches, and cyber threats.

2. Common Structural Vulnerabilities:

- SQL Injection: Attackers manipulate database queries through unsanitized input fields.
- Cross-Site Scripting (XSS): Malicious scripts injected into web pages affecting user sessions.
- Broken Authentication: Weak session management allowing unauthorized access.

3. OWASP ZAP Security Validation Workflow:

Target Configuration: Configure demo application URL inside OWASP ZAP.

Passive Scanning: Automatically analyzes HTTP traffic without attacking the system.

Active Scanning: Performs controlled attacks to identify vulnerabilities.

Vulnerability Identification: ZAP lists detected issues such as XSS or Injection flaws.

Severity Classification: Vulnerabilities categorized as High, Medium, Low, or Informational.

Evidence Documentation: Generate HTML/PDF scan reports including screenshots and logs.

Vulnerability Findings Table

SQL Injection	Database query manipulation	High	Use prepared statements
Cross-Site Scripting	Script injection in web pages	Medium	Input validation & encoding
Broken Authentication	Session hijacking risk	High	Strong authentication controls

Section D – Validation Risk Structuring and Compliance Evidence Compilation

1. Validation Risk Register

Risk	Impact	Likelihood	Mitigation Strategy
Unauthorized Access	Patient data breach	High	Multi-factor authentication
Data Loss	Loss of medical records	Medium	Automated backups
Unvalidated Input	System compromise	High	Input validation & sanitization

2. Compliance Evidence Dossier Structure

A compliance evidence dossier organizes validation proof for audits.

Recommended GitHub/Docs Structure:

- /validation-checklist/
- /ci-workflows/
- /security-reports/
- /risk-register/
- README.md explaining validation process

This structure ensures traceability, audit readiness, and regulatory compliance.

Submission Requirements

Upload:

- Validation checklist document
- GitHub Actions YAML workflow
- OWASP ZAP scan evidence
- Risk register document

Submit GitHub repository link containing implementation evidence and reports.